

LESSON 6-1 • HOMEWORK PACKET • SHEET 1 OF 2**Dividing by a Unit Fraction**

Maryland Standard 6.NS.A.1 · Learning target – I can divide a whole number by a unit fraction and explain what the quotient counts.

REMEMBER

- Dividing by a **unit fraction** asks: how many pieces of that size fit into the number?
- 1 whole holds 4 fourths, so $3 \div \frac{1}{4} = 3 \times 4 = 12$.
- The quotient is **greater** than the number you divide.
Check: $12 \times \frac{1}{4} = 3$.

WORKED EXAMPLE

How many $\frac{1}{3}$ -cup scoops are in 5 cups of rice?

Step 1 Each cup holds 3 thirds.

Step 2 5 cups hold $5 \times 3 = 15$ thirds, so $5 \div \frac{1}{3} = 15$.

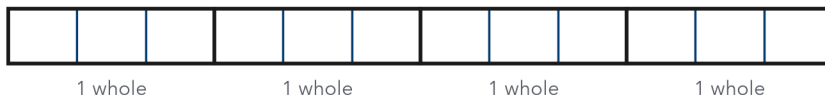
Step 3 Check: $15 \times \frac{1}{3} = 5$, so it checks.

Answer 15 scoops

PRACTICE

show your work · label units

1. Each bar is 1 whole. Use the fraction bars to find $4 \div \frac{1}{3}$. How many thirds are in 4 wholes?



$4 \div \frac{1}{3} =$ _____

2. Divide.

a. $6 \div \frac{1}{2} =$ _____

b. $3 \div \frac{1}{5} =$ _____

c. $8 \div \frac{1}{4} =$ _____

d. $2 \div \frac{1}{9} =$ _____

e. $7 \div \frac{1}{3} =$ _____

f. $10 \div \frac{1}{6} =$ _____

g. $5 \div \frac{1}{7} =$ _____

h. $12 \div \frac{1}{3} =$ _____

3. **Trail Mix.** Jamal has 6 cups of trail mix. He fills snack bags that each hold $\frac{1}{4}$ cup. How many bags can he fill?

Answer _____ bags

4. **Ribbon Bows.** Ms. Carter has 5 yards of ribbon. She cuts it into pieces that are each $\frac{1}{6}$ yard long. She uses 18 pieces to make bows. How many pieces are left?

Show your work.

Answer _____ pieces

5. **Find the Error.** Tanya says $8 \div \frac{1}{2} = 4$. Explain her mistake. Then find the correct quotient.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Count how many $\frac{1}{4}$ -cup scoops fill 2 cups. Does it match $2 \div \frac{1}{4}$?

LESSON 6-1 • HOMEWORK PACKET • SHEET 2 OF 2**Dividing a Fraction by a Whole Number**

Maryland Standard 6.NS.A.1 · Learning target – I can divide a fraction by a whole number and explain the result.

REMEMBER

- Dividing by a whole number **shares** a fraction into equal parts.
- Multiply by the reciprocal: $\frac{2}{3} \div 4 = \frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$.
- Dividing by a whole number greater than 1 gives a quotient **less** than the fraction. Check: $\frac{1}{6} \times 4 = \frac{2}{3}$.

WORKED EXAMPLE3 friends share $\frac{3}{4}$ of a sandwich equally. What part of the whole does each get?**Step 1** $\frac{3}{4} \div 3 = \frac{3}{4} \times \frac{1}{3} = \frac{3}{12}$ **Step 2** Simplify: $\frac{3}{12} = \frac{1}{4}$ **Answer** $\frac{1}{4}$ of the sandwich**PRACTICE***show your work · label units*

1. Divide. Write each quotient in simplest form.

a. $\frac{1}{2} \div 3 =$ _____

b. $\frac{4}{5} \div 2 =$ _____

c. $\frac{3}{8} \div 3 =$ _____

d. $\frac{2}{3} \div 6 =$ _____

e. $\frac{6}{7} \div 2 =$ _____

f. $\frac{5}{6} \div 10 =$ _____

2. Find each share. Write it in simplest form.

a. 4 people share $\frac{1}{2}$ lb of cheese: _____ lb

b. 5 cups share $\frac{5}{6}$ L of juice: _____ L

- 3.
- Paint Cans.**
- Kiara has
- $\frac{3}{5}$
- gallon of paint. She pours it equally into 6 small cans. How much paint is in each can?

*Show your work.***Answer** _____ gallon

- 4.
- Clay.**
- Mr. Okafor has
- $\frac{2}{3}$
- pound of clay. He splits it equally among 4 students. How much clay does each student get?

Answer _____ pound

- 5.
- Reason.**
- Without dividing, decide: is
- $\frac{5}{8} \div 4$
- greater than or less than
- $\frac{5}{8}$
- ?

☐ greater than $\frac{5}{8}$ ☐ less than $\frac{5}{8}$ **TEST PRACTICE***choose the best answer*

6. Which expression has the same value as
- $\frac{5}{6} \div 5$
- ?

A $\frac{5}{6} \times 5$

B $\frac{5}{6} \times \frac{1}{5}$

C $\frac{6}{5} \times 5$

D $\frac{5}{6} - 5$

7. 4 people share
- $\frac{4}{9}$
- of a pizza equally. What fraction of the whole pizza does each get?

A $\frac{1}{9}$

B $\frac{16}{9}$

C $\frac{1}{36}$

D $\frac{4}{9}$

CHECK YOURSELF☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** Share $\frac{1}{2}$ of a sandwich equally among 2 people at home. What fraction of the whole sandwich does each get?